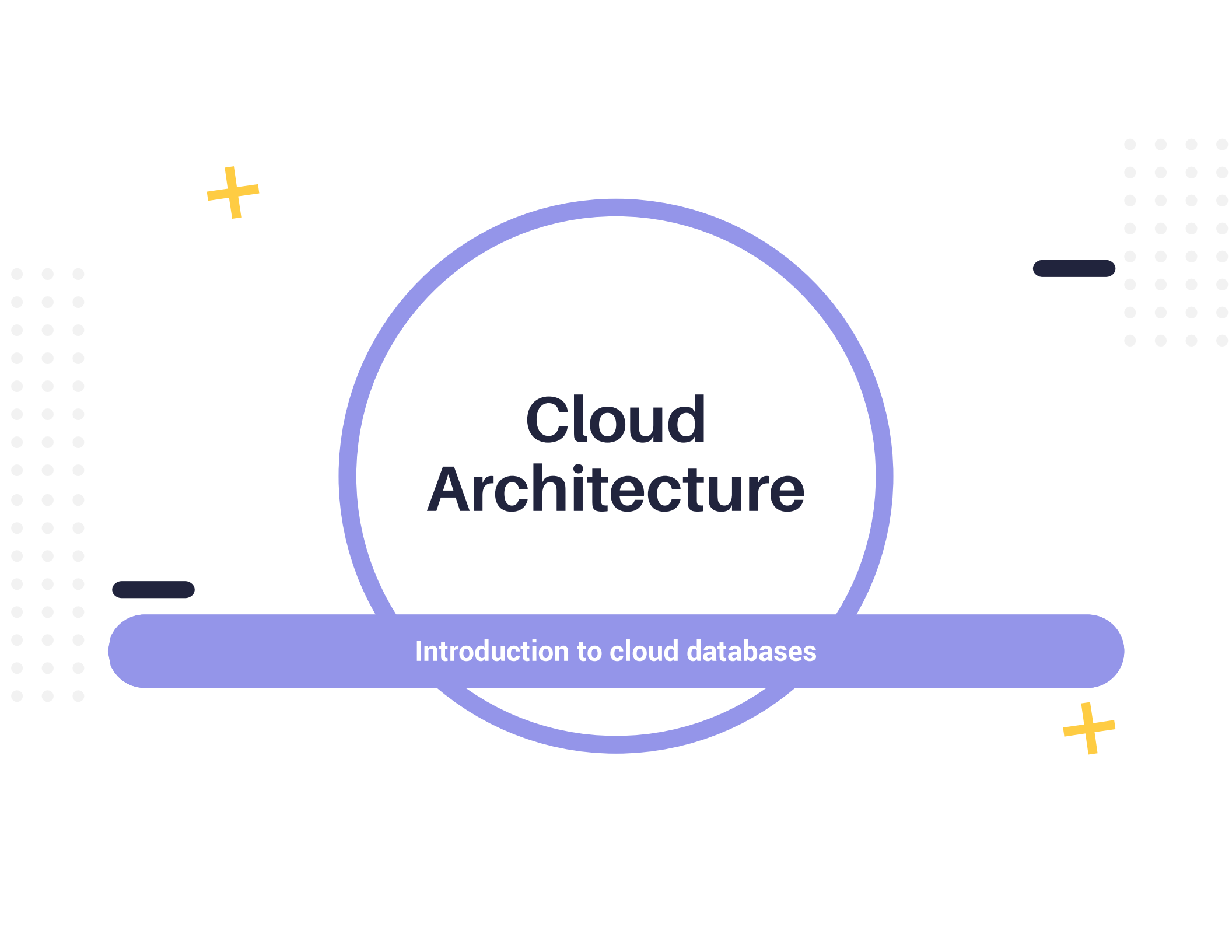




Cloud Architecture

Introduction to cloud databases

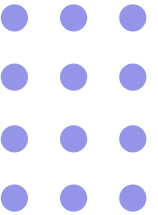


Objectives

Define relational
databases

Explore database
management

Create Oracle APEX
environment



What is a relational database?



Relational database

A type of database that organises data into one or more tables of columns and rows, with a unique identifier for each row.

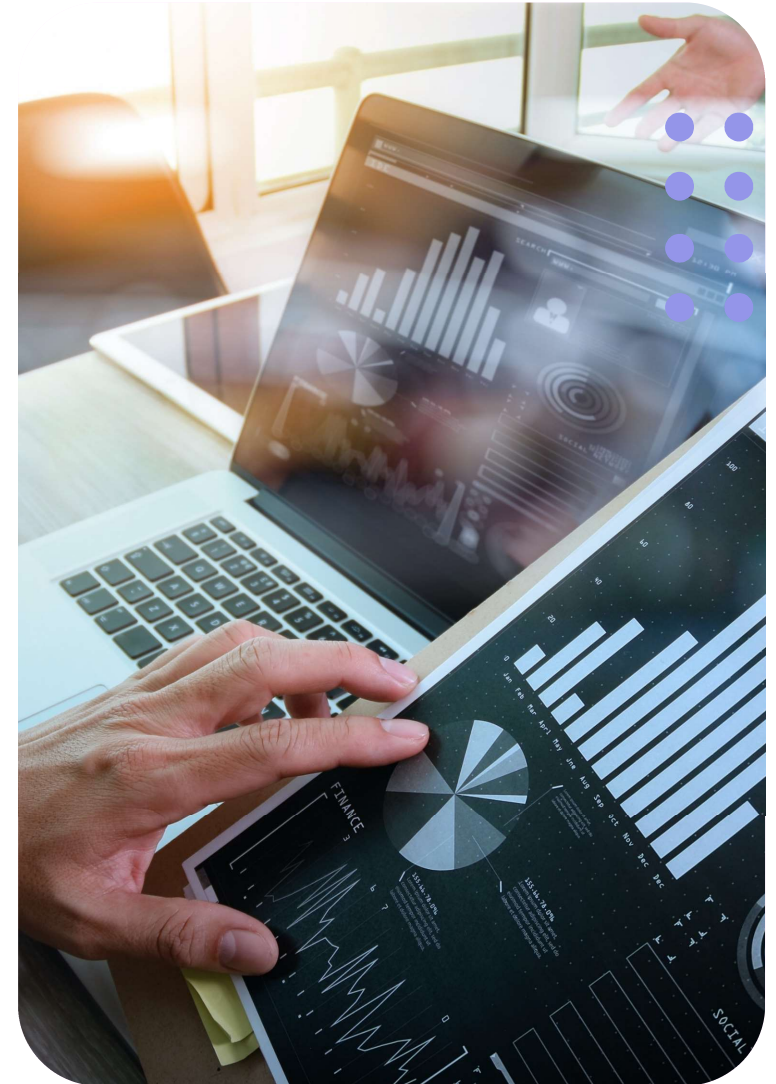


Characteristics of relational databases

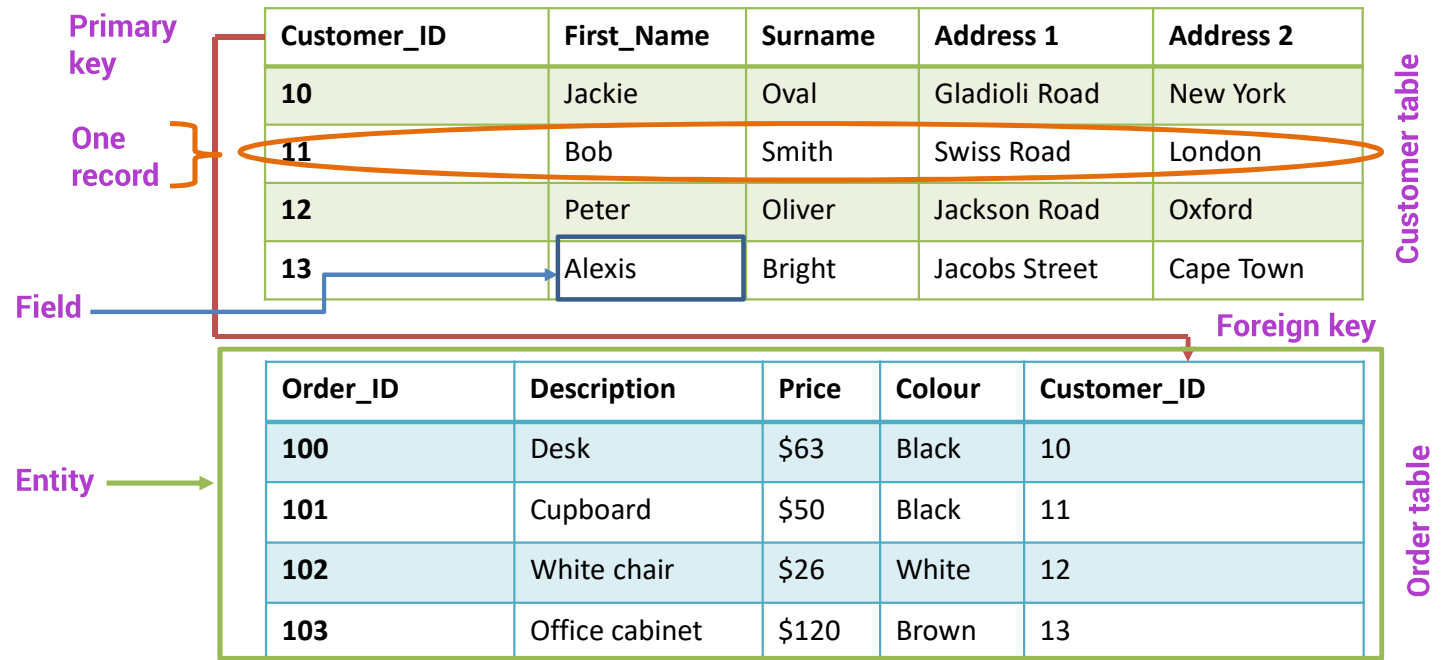
Consists of tables

Makes use of records and fields

Constrains data using primary and foreign keys



Characteristics of relational databases



**Relational database
management
systems (RDBMS)**



Relational database management systems (RDBMS)

A RDBMS is a software system that is used to manage and interact with a relational database.



Considerations for developing a RDBMS

Content

Access

Physical organisation

Logical structure



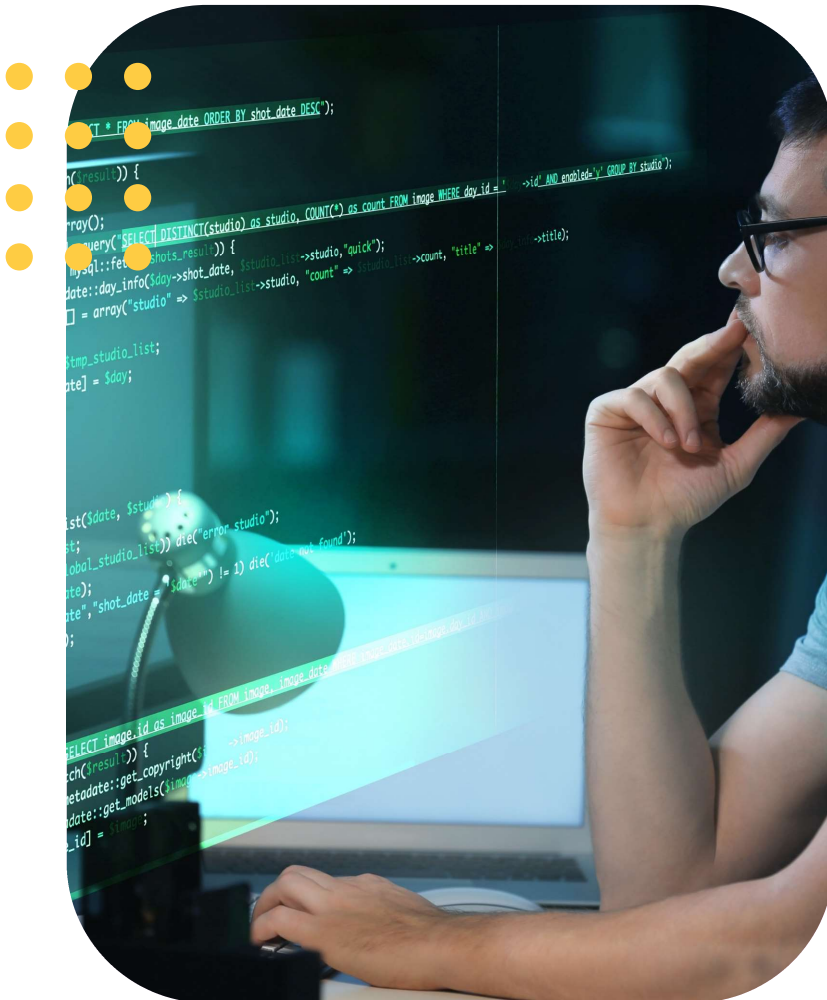
Role of a database administrator (DBA)

Define the data needs of users

Identify entities of interest

Applies database programming languages

Ensures data is secure from unauthorised access





Types of database management systems

Single-user

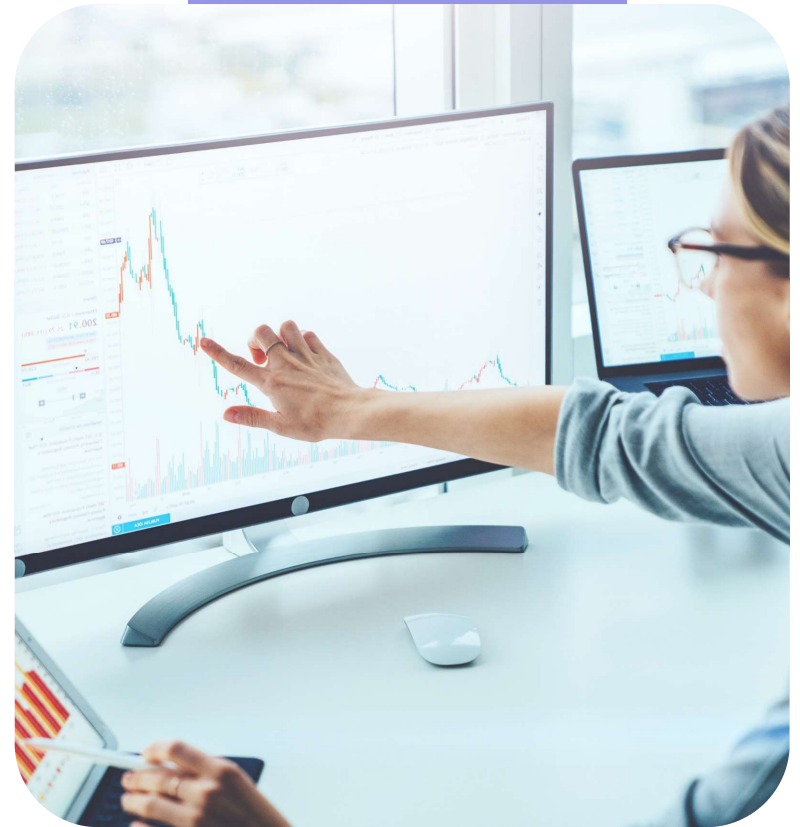
Multi-user

Flat files



SQL database programming

Structured Query Language (SQL) is a special-purpose programming language for accessing and manipulating data stored in a relational database.



Features of SQL databases

Select data

Project data

Join tables





Types of SQL database management systems

MySQL

Oracle
Database

MS SQL
Server



Getting started with Oracle APEX

DEMO HERE